Al-Isra university
Faculty of Engineering
Electrical Engineering Department
Eng. Juma Abu-Ghalune
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Second Exam
Signals and Systems Analysis

1. Find the Fourier Transform of $x(t) = exp(-4t).u(t - 2)$ using the integral form.

2. The complex exponential Fourier Series over an interval $0 < t < T$ is $x(t) = \sum (4/(1 + j\pi n))exp(-j20nt/4)$ for all n. Find:
   a. The average value. b. The fundamental frequency.
   c. The amplitude and the phase of the third harmonic.

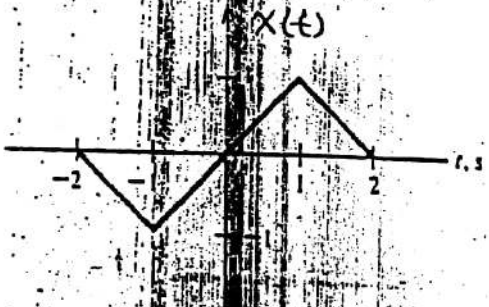
3. Using Table 4-1 and Table 4-2, find the Fourier Transform inverse of:
   a. $X(f) = (1/f)exp(j20\pi f)$.. b. $Y(f) = 4/|2 + j\pi f|^2$.
   c. $Z(f) = 4\pi f.sinc^2(2f)$.

4. Use Parseval's theorem to find the average power for the signal $x(t) = 4.sin^2(150\pi t).cos(200\pi t)$.

5. Consider a fixed linear system with amplitude response $2A - A.\Pi[f/200]$ and phase response $-0.01\pi f$ for $|f| < 50$, 0.5π for $f < -50$, and -0.5π for $f > 50$. Find the output and determine the type of distortion for the following inputs:
   a. $y(t) = cos(120\pi t) + cos(220\pi t)$. b. $w(t) = cos(80\pi t) + 4\,cos(160\pi t)$.

6. Use the differentiation theorem to find the Fourier Transform of the signal shown in Fig. 1.

Fig. 1



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Q.1 $x(t) = e^{-4t} u(t-2)$

$$X(f) = \int_2^{\infty} e^{-(j2\pi f + 4)t}\, dt = \int_{+2}^{\infty} e^{-t(4+j2\pi f)}\, dt$$

$$= \frac{-1}{(4+j2\pi f)} e^{-t(4+j2\pi f)} \Big]_{+2}^{\infty}$$

$$= \frac{1}{(4+j2\pi f)} e^{-2(4+j2\pi f)} = \frac{e^{-8}}{(4+j2\pi f)} e^{-j4\pi f}$$

Q.2 $x(t) = \sum_{n=-\infty}^{\infty} \frac{4}{(1+j\pi n)} \cdot e^{j5nt} \qquad 0 < t < T$

a. $n = 0 \rightarrow X_0 = 4$: average value

b. $n = 1 \rightarrow 2\pi f_0 = 5 \rightarrow f_0 = \frac{5}{2\pi}$

c. $X_3 = \frac{4}{(1+j3\pi)} = \frac{4}{\sqrt{1+9\pi^2}} \angle -\tan^{-1}(3\pi)$

$$= \frac{4}{\sqrt{1+9\pi^2}} \angle -83.9$$

Q3 a. $X(f) = \frac{1}{f} e^{j20\pi f} \longleftrightarrow j\pi \operatorname{sgn}(t+10)$

Time delay + 14.

b. $Y(f) = \frac{4}{|2+j\pi f|^2} = \frac{4}{4+(\pi f)^2} = \frac{16}{16+(2\pi f)^2} = 2\frac{2(4)}{4^2+(2\pi f)^2}$

$y(t) = 2e^{-4|t|}$

c. $Z(f) = 4\pi f \operatorname{sinc}^2(2f) = -j(j2\pi f)\, 2\operatorname{sinc}^2(2f)$.

$$\frac{d\Lambda\left(\frac{t}{2}\right)}{dt} \quad -j\pi\left(-\frac{1}{2}\right) \quad +j\pi\frac{1}{2}$$

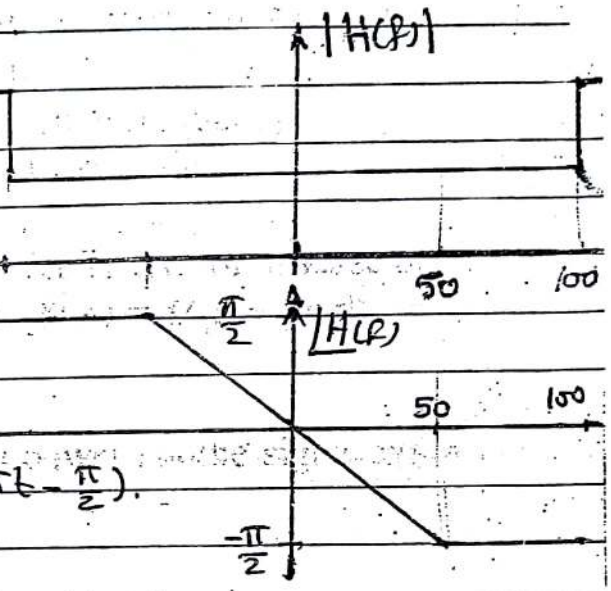
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24 $x(t) = 4\sin^2(150\pi t)\cos(200\pi t) = 2[1-\cos(300\pi t)]\cos(200\pi t)$

$= -\cos(100\pi t) + 2\cos(200\pi t) + \cos(500\pi t)$

$X_1 = \frac{-1}{2}, \quad X_2 = 1, \quad X_5 = \frac{-1}{2}$

$P_{avg} = 2\left[\frac{1}{4} + 1 + \frac{1}{4}\right] = 3$



25 a. $y(t) = \cos(120\pi t) + \cos(220\pi t)$

$y_o(t) = A\cos(120\pi t - \frac{\pi}{2}) + 2A\cos(220\pi t - \frac{\pi}{2})$

Amplitude distortion.

b. $w(t) = \cos(80\pi t) + 4\cos(160\pi t)$

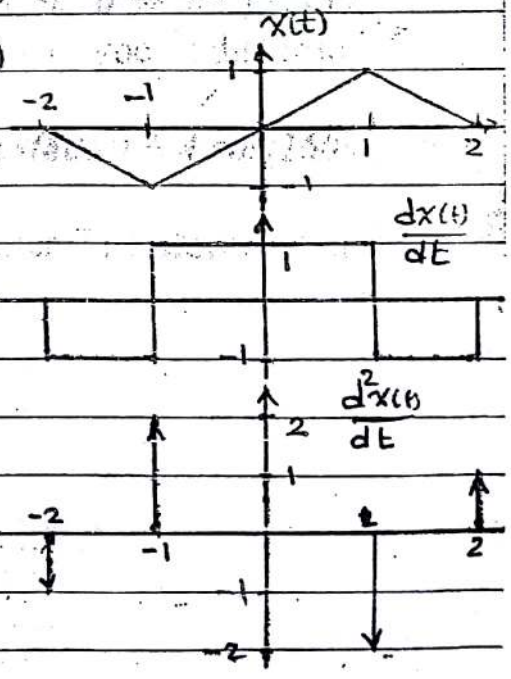
$w_o(t) = A\cos(8\pi t - 0.4\pi) + 4A\cos(160\pi t - \frac{\pi}{2})$.

Phase distortion.

26 $\frac{d^2x(t)}{dt^2} = +\delta(t-2) + \delta(t+2) - 2\delta(t-1) + 2\delta(t+1)$

$F\{\frac{d^2x(t)}{dt^2}\} = e^{-j2\pi f2} + e^{j2\pi f2} - 2e^{-j2\pi f} + 2e^{+j2\pi f}$

$= -j2\sin(4\pi f) - j4\sin(2\pi f)$.



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